

A 1512 PPI Real RGB Glassed-OLED Display for VR

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Abstract

This paper presents a 2.56-inch Real RGB glass-based OLED display with a pixel density of 1512 PPI. To address critical manufacturing technical challenges—such as the intricate patterning of fine metal lines, stress compatibility in multi-layer film stacking, and circuit reliability issues caused by complex topographies—a systematic process integration strategy is proposed, incorporating ACD (Array Circuit Design) OPC (Optical Proximity Correction) compensation, film stress optimization, and trench filling processes. The resulting display achieves an ultra-high resolution of 2560×2740 while significantly reducing manufacturing costs. It also offers comprehensive advantages including a 120Hz high refresh rate, a 1,000,000:1 contrast ratio, and low power consumption. This product provides key technical support and commercial impetus to overcome the bottlenecks in VR display technology and promote industrial scaling.

Author Keywords

Glass base OLED; Real RGB; ultra-high PPI; VR; AR.

1. Introduction

In recent years, virtual reality (VR) and augmented reality (AR), as the next-generation human-computer interaction platforms, have seen their near-eye display markets experience rapid growth, demonstrating huge application potential. The quality of immersive experiences largely depends on the performance of near-eye display devices. Among them, high pixel density, low latency, low power consumption and cost-effectiveness are the keys to the large-scale popularization of VR/AR devices [1, 2]. To meet the demand for ultra-high pixel density, the industry has long mainly explored two technical paths: the traditional Fast-LCD solution represented by fast-response liquid crystal, and the cutting-edge display solution represented by Micro-OLED.

Fast-LCD (Fast Liquid Crystal) technology, with its mature industrial chain and low-cost advantage, is widely applied in the mid-to-low-end market. However, its inherent characteristics have led to many performance compromises, including relatively slow response time (which can easily cause image trailing), high power consumption, and being far inferior to self-luminous OLED technology in key image quality indicators (such as contrast and color gamut), failing to meet the requirements of professional-level applications for immersion and visual fidelity [3].

Micro-OLED (silicon-based OLED) technology excels in performance, capable of achieving ultra-high pixel density, high contrast and rapid response, making it the preferred choice for high-end VR devices. However, its manufacturing process relies on expensive semiconductor silicon wafers as the driving backplane, which not only has high material costs but also a complex manufacturing process, facing yield challenges, resulting

in persistently high terminal product prices. This has become the core bottleneck restricting the large-scale development of the high-end VR industry [4, 5].

Against this backdrop, we propose and focus on analyzing a highly promising third-generation solution - Glass-based OLED (G-OLED) display technology. Take the latest released 2.56-inch Real RGB G-OLED product in this study as an example. This technology has successfully achieved the world's highest pixel density of 1512 PPI on a glass substrate, and its resolution is as high as $2560 \text{ (RGB)} \times 2740$. Compared with traditional solutions, GOLED demonstrates significant comprehensive advantages: In terms of performance, it not only meets the 1512 PPI ultra-high resolution required by professional VR displays, but also features an ultra-high contrast ratio of 1,000,000:1, a 120Hz high refresh rate, and lower power consumption, effectively enhancing device battery life and reducing visual fatigue. In terms of economy, the mature glass substrate process it adopts fundamentally avoids the cost pain point of silicon-based OLED, providing a cost-effective solution for achieving high-performance VR displays.

2. Experiments and Methods

2.1 Ultra-high PPI Array design

With the increasing demand for high PPI, the backplane array design has put forward extreme requirements for line width/line spacing (current target: $L+S \leq 2.8\mu\text{m}$). This specification is critical for ensuring the electrical performance of metal lines—such as meeting resistance requirements while simultaneously eliminating the risk of short circuits between different signal lines. However, under the physical limitations of the existing exposure machine and photomasks, achieving such fine and high-density circuit patterns simultaneously poses a huge challenge and is highly likely to cause defects such as open circuits or short circuits. To overcome these challenges, we have successfully achieved a high-yield graphical process through an integrated technical solution that combines mask design, optical compensation and process optimization.

1. Enhance mask specifications and introduce ACD OPC compensation: By adopting higher-specification photomasks and applying advanced optical proximity effect correction technology, pre-distortion compensation for the pattern is carried out during the mask design stage. This effectively suppresses the graphic distortion caused by light diffraction and interference effects, ensuring the design fidelity of the target graphic from the source.

2. Collaborative optimization of exposure and etching processes: At the process level, we precisely adjust the exposure conditions for photolithography to achieve the best post-development graphic profile. Meanwhile, the subsequent etching process parameters are collaboratively optimized to ensure that the patterns on the

photoresist can be precisely transferred to the underlying film layer, ultimately obtaining metal circuits with clear contours and accurate dimensions.

Through the synergistic effect of the above measures, we have successfully achieved ultra-high-density and micro-sized circuit graphics. All achievements meet the product specification requirements and effectively eliminate open circuit and short circuit problems, laying a solid foundation for the manufacturing of ultra-high PPI display screens.

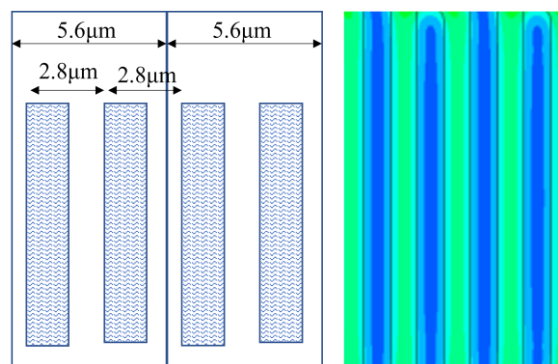


Figure 1. Schematic diagram of array design

2.2 Research on Stress Mismatch of Film Layers

To meet the multi-TFT driving architecture requirements of ultra-high PPI backplanes, the number of signal lines has doubled, the size of vias has been significantly reduced, and their density is extremely high. To this end, we adopted a composite insulation layer architecture to suppress signal interference and parasitic capacitance. However, this complex multi-layer stacking structure, especially the alternating stacking between inorganic and organic film layers, due to the differences in material properties, is highly prone to a decline in interfacial bonding force during the manufacturing process due to mismatched thermal expansion coefficients and internal stress accumulation, thereby causing fatal defects such as film layer peeling or bulging. As shown in Fig. 2, it is a schematic diagram of the peeling SEM(a) plane (b) section of the film layer and the SEM(c) plane (d) section of the bulge. To address this severe reliability challenge, we focused on the characteristics of the film layer itself and the film-forming process. By optimizing the film-forming process and improving the film quality, we achieved proper stress matching between layers and improved the Peeling/ bulging problem.

1. Optimization of film quality and film-forming process: By systematically adjusting the core process parameters such as the chemical vapor deposition or coating formula, deposition temperature, pressure and power for each film layer (especially the key inorganic insulating layer and organic flat layer), we precisely controlled the intrinsic stress and mechanical properties of each film layer.

2. Achieve stress matching design: The above optimization aims to make the stress state (compressive stress or tensile stress) and magnitude of adjacent film layers in the stacked structure mutually compatible, effectively reducing the net stress of the entire multi-layer and thereby significantly weakening the driving force causing peeling and bulging.

Through a stress-engineering-centric technical approach, we successfully realized the excellent interfacial bonding force between the film layers, basically eliminated the peeling and bulging problems, and finally obtained the backplane with stable

structure and high reliability.

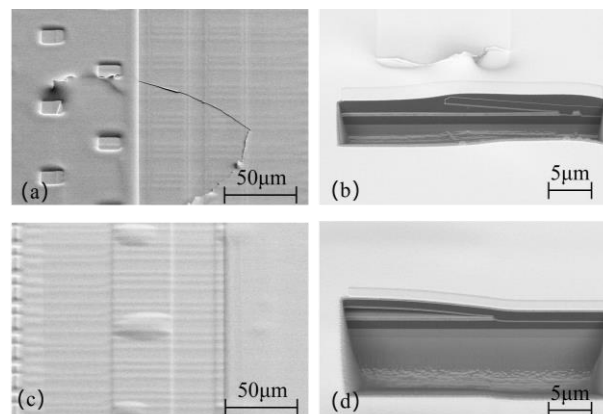


Figure 2. Peeling SEM(a) plane (b) section of the film layer, SEM(c) plane (d) section schematic diagram of the bulge

2.3 Research on Stress Mismatch of Film Layers

In ultra-high PPI backplanes, the multi-TFT drive architecture and ultra-high density graphics result in extremely complex three-dimensional array terrains. This significant undulation poses a huge challenge to the upper layer of metal in the photolithography and etching processes: uneven coverage of photoresist at the steps, as well as residue or over-etching during the etching process, jointly lead to a high incidence of open circuits and short circuits in metal circuits, seriously threatening the yield and reliability of the circuits. To overcome this difficult problem, we introduced the planarization layer in the Array process to fill the uneven terrain areas within AA. It is easier to carry out the complex pattern manufacturing process based on the flattened terrain.

1. Active topographic planarization: By depositing dedicated planarization layers before the formation of metal layers, uneven terrain areas and steps in the active zone are actively filled. This move greatly improved the overall flatness of the surface.

2. Ensuring process Windows and electrical performance: The subsequent production of complex patterns on a flattened surface significantly broadens the process Windows for photolithography and etching, fundamentally reducing the risk of open circuits and short circuits caused by undulating terrain.

Meanwhile, the flat base ensures the uniformity and consistency of the signal line resistance and capacitance, guaranteeing the electrical performance of the design.

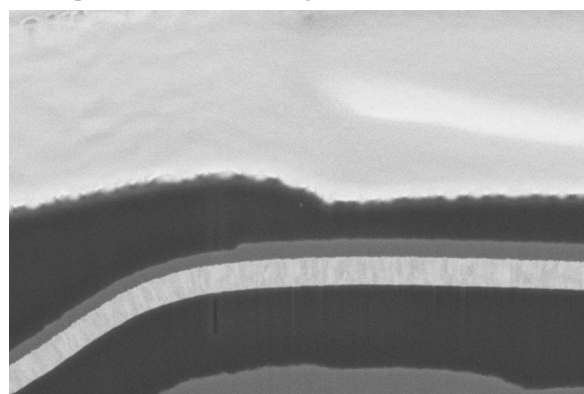


Figure 3. Schematic diagram of complex terrain

3. Results and Discussion

The 1512 PPI GOLED display to be showcased in 2025 marks a major breakthrough for glass-based technology in the field of ultra-high-resolution near-eye displays. The system architecture of this product innovatively integrates high-performance pixel circuit design, optimized driver schematic diagram, efficient top-emitting OLED devices, and combines advanced packaging technology with Real RGB colorization solution, jointly providing key display support for the next generation of VR/AR devices. Table 1 lists the main content specifications of the 2.56-inch GOLED panel.

Table 1. 2.56-inch GOLED Panel Specification

Spec. Items	Specification
Display Size	2.56 inch
Resolution(Real RGB)	2560(RGB)*2740
PPI	1512
Max Frame Rate	120Hz
Contrast Ratio	>1000000:1
Duty Function	Supported
Backplane	Glass

Fig. 4(a) shows the stacked OLED emitting red with chromaticity coordinates of CIE 1931 $(x, y) = (0.657, 0.333)$, and Fig. 4(b) shows the same OLED emitting blue with CIE 1931 $(x, y) = (0.170, 0.156)$, and Fig. 4(c) shows the OLED emitting green with CIE 1931 $(x, y) = (0.232, 0.663)$, and Fig. 4(d) shows the OLED emitting white with CIE 1931 $(x, y) = (0.400, 0.393)$. The white luminescence of this device is achieved by RGB color mixing.

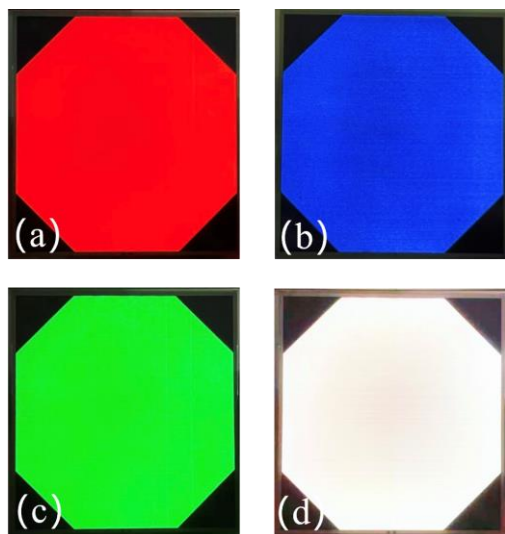


Figure 4. Emitting (a) Red (b) Blue (c) Green (d) White

In response to the core pain points of the popularization of the VR industry, TCL CSOT will simultaneously release a 2.56-inch real RGB glass-based OLED product in 2025. With the dual advantages of high resolution and low cost, it will inject key impetus into the large-scale development of the VR industry. The world's highest PPI Real RGB glass-based OLED display (2.56"). As shown in Figure 5, it is the overall effect drawing of the

product at the exhibition, the partial effect drawing of the video playback (b), and the display picture (c).

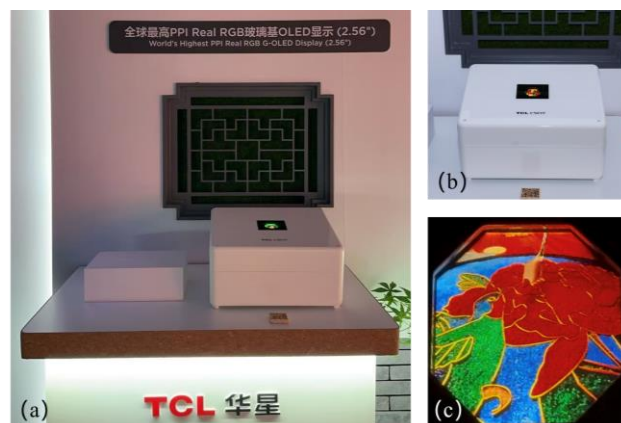


Figure 5. Overall effect drawing of the product (a), partial effect drawing of video playback (b), display picture (c) at the 2025 TCL CSOT DTC exhibition

4. Discussion

GOLED has demonstrated significant comprehensive advantages in the field of VR/AR near-eye display. Compared with the high cost of silicon-based OLED and the performance limitations of fast-LCD, GOLED, by innovatively adopting glass substrates, achieve outstanding performance such as an ultra-high resolution of 1512 PPI, a refresh rate of 120Hz, and a contrast ratio of millions, while significantly reducing manufacturing costs. This breakthrough technology effectively addresses the core pain points faced by the VR industry during its popularization process, providing an ideal display solution for the large-scale application of terminal devices.

5. References

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